

MUHAMMAD ASAD

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PROFESSIONAL SUMMARY

Software Engineering student and backend development trainee with hands-on experience building REST APIs using Django REST Framework, PostgreSQL, and JWT-based authentication. Strong foundation in Data Structures & Algorithms, OOP design, and software architecture patterns, with a track record of independently shipping 10+ full-stack and algorithmic projects.

EDUCATION

Bachelor of Engineering in Computer Software Engineering

Bahria University – Karachi Campus

Sep 2024 – Sep 2028

Intermediate – Computer Science (FSc)

Malir Cantt College

Completed

EXPERIENCE

Backend Development Trainee (Hybrid)

ByteCorp Technologies – Training Program

Jul 2026 – Present (3-Month Program)

- Collaborating with a technical mentor on the end-to-end development of a production-style backend system, from database design through API implementation and security.
- Following a structured PR-based workflow, submitting code for review and iterating on mentor feedback to meet production-grade engineering standards.
- Taking ownership of independent modules, including schema design, authentication, and authorization, while adhering to industry-standard Git and version control practices.
- Delivered Phase 1 of the project on schedule, with subsequent phases planned to expand into advanced software design patterns.

Software Development Intern (Remote)

CodeAlpha – Project-Based Internship

2024 (4-Week Engagement)

- Delivered two production-quality Java applications (Student Grade Manager, Hotel Reservation System) under real internship specifications, both reviewed on GitHub.
- Applied clean code principles, OOP design patterns, Git version control, and project documentation throughout the engagement.
- Both submissions were adopted as reference implementations on the internship platform following positive architecture review.

PROJECTS

FoodHeaven – Restaurant POS System | Python, Django REST Framework, PostgreSQL, JavaScript, PyWebView

- Built a native desktop Point of Sale system for restaurants with a dark-themed register UI, real-time cart management, GST calculation, and a Kitchen Monitor Display for live order tracking.
- Developed an Analytics Dashboard using Chart.js for sales trends, along with printable invoice generation and multi-mode checkout backed by a PostgreSQL database across modular Django apps.

Job Ranking System (SkillMesh) | Python, DSA, Greedy Algorithms, OOP

- Built a DSA-based candidate ranking and recommendation engine that scores applicants using skill matching, experience scoring, and greedy budget optimization algorithms.
- Implemented a Max Heap for efficient top-candidate retrieval, with an Explain Mode for step-by-step algorithm tracing.

Requirement Genie | Python, NLP, AI/ML, Streamlit

- Developed an AI-driven requirements engineering assistant that automates generation of SRS (Software Requirements Specification) documents using NLP techniques.
- Integrated a conversational interface for systematic requirements gathering, reducing manual documentation effort.

Algo Puzzle Board | HTML, CSS, JavaScript, Data Structures & Algorithms

- Engineered an interactive algorithm visualization platform covering sorting, pathfinding, N-Queens, Knight's Tour, Graph Coloring, TSP, Huffman Coding, Dijkstra's, and Kruskal's MST.

- Designed a modular dashboard architecture with animated step-by-step algorithm walkthroughs to aid conceptual understanding.

Food Heaven Website | HTML, CSS, JavaScript, Bootstrap 5, .NET Framework

- Built a restaurant management website with a modern UI, dynamic menu display, and a table reservation system with responsive design.
- Implemented client-side interactivity using vanilla JavaScript and Bootstrap 5 grid for a seamless experience across devices.

MIPS Productivity Dashboard | MIPS Assembly, Computer Architecture

- Built a low-level task management system entirely in MIPS Assembly, featuring task tracking, calendar view, and direct memory status updates.
- Demonstrated command of computer architecture, register management, and memory-mapped I/O in a resource-constrained environment.

Additional projects (Hotel Reservation System, Decode & Discover, Streamlit Dashboard Suite, and 10+ Python/Java mini-projects) available on GitHub and Portfolio.

TECHNICAL SKILLS

Languages: C#, Java, Python, JavaScript, MIPS Assembly, SQL, HTML, CSS

Frameworks & Libraries: Django, Django REST Framework, Bootstrap 5, Streamlit, Pandas, Matplotlib, NumPy, Java Swing, C# Windows Forms, PyWebView, Chart.js

Databases: PostgreSQL, MySQL, SQL Server, SQLite

Developer Tools: Git, GitHub, VS Code, Visual Studio, IntelliJ IDEA, PyCharm, Netlify

Core Concepts: OOP, Data Structures & Algorithms, Design and Analysis of Algorithms, REST APIs, NLP, Computer Architecture, Agile Basics

Software Architecture & Design: Design Patterns, Design Principles (SOLID), MVC, Client-Server, Layered Architecture, Peer-to-Peer, Pipe-and-Filter

ACHIEVEMENTS & HIGHLIGHTS

- 14+ open-source projects on GitHub across Java, Python, C#, and JavaScript with a consistent commit history demonstrating real-world development discipline.
- Portfolio deployed on Netlify showcasing 10+ production-quality projects spanning web development, AI tools, algorithm visualizations, and systems programming.
- CodeAlpha Internship: completed a remote software development internship, delivering enterprise-style Java applications under real project deadlines.
- Self-directed learning: independently mastered Streamlit, Pandas, NLP basics, and data visualization tools beyond coursework; shipped multiple deployable applications.
- Built an AI-powered requirements engineering tool (Requirement Genie), demonstrating initiative in applying NLP to real software engineering problems.